The issue why this breaks is the following case: Assume for two oracles $\mathcal{O}_j, \mathcal{O}_k$ there exist obscure queries and both are secure vrf oracles. Then we cannot expect that by branching over both sets of obscure oracle queries that the output for the true secret key sk^* is y over all branches.

That is because the obscure queries of a secure vrf oracle $\mathcal{O}_i$ are only oracle-wide jointly-irrelevant. If we would follow the proof over two changed oracles $\mathcal{O}_j, \mathcal{O}_k$ then we'd arrive at a point where we would need to consider the second execution which is possibly not uniquely provable or perfectly correct and therefore violate the robustness condition, breaking the contradiction.

We *still* know that oracle-wide joint-independence holds however. Assume f_t is the PPT Turing machine in the terminal node and fix the inputs $\mathbf{y}$ corresponding to oracle outputs to *exposed* queries of $\mathcal{O}_{vrf}$ or queries to $\mathcal{O}_{uppc}$. Then if we group the remaining *obscure* queries to $\mathcal{O}_{vrf}$ by oracle:

$$f_t(\underbrace{\mathbf{C}}_{\text{known queries}}, \underbrace{\mathbf{y}_1^{(1)}, \ldots, \mathbf{y}_{k_1}^{(1)}}_{\text{obscure queries to } \mathcal{O}_1}, \ldots, \underbrace{\mathbf{y}_1^{(n)}, \ldots, \mathbf{y}_{k_n}^{(n)}}_{\text{obscure queries to } \mathcal{O}_n}) = f_t(\mathbf{C}, \vec{\mathbf{y}}^{(1)}, \ldots, \vec{\mathbf{y}}^{(n)})$$

We know that, by oracle-wide joint irrelevance, f_t must satisfy:

$$\forall k \in [n], \forall j \in [n], \forall \vec{\mathbf{y}}^{(j)}, \vec{\mathbf{z}}^{(k)} \text{ well formed}, f_t(\mathbf{C}, \vec{\mathbf{y}}^{(1)}, \ldots, \vec{\mathbf{y}}^{(k)}, \ldots, \vec{\mathbf{y}}^{(n)}) = f_t(\mathbf{C}, \vec{\mathbf{y}}^{(1)}, \ldots, \vec{\mathbf{z}}^{(k)}, \ldots, \vec{\mathbf{y}}^{(n)})$$

We say the function corresponding to f_t is *axis-constant* since for every single possible input (with fixed $\mathbf{C}$) the output remains constant if we keep at least $n-1$ inputs constant. If there doesn't exist a robust $(2;3)$ black-box VRF-combiner than this means that there exist two input values such that keeping the third constant, there exist two different, well-formed assignments which results in a different output. That is:

$$\exists k_1, k_2 \in [3], \forall j \in [3], \forall \vec{\mathbf{y}}^{(j)}, \vec{\mathbf{z}}^{(k_1)}, \vec{\mathbf{z}}^{(k_2)} \text{ well formed}, f_t(\mathbf{C}, \vec{\mathbf{y}}^{(j)}, \vec{\mathbf{y}}^{(k_1)}, \vec{\mathbf{y}}^{(k_2)}) \neq f_t(\mathbf{C}, \vec{\mathbf{y}}^{(j)}, \vec{\mathbf{z}}^{(k_1)}, \vec{\mathbf{z}}^{(k_2)})$$

The idea. Only consider branches that can be jointly embedded into a globally consistent family of secure VRFs. We show that in that case axis-irrelevance must generalize to global joint irrelevance of the obscure oracle queries. Specifically when branching the simulator $\mathsf{EvalSim}_{\mathsf{VRF}}$ maintains a list of objects $(\mathsf{sk}, \mathsf{pk})$ and mappings $(\mathsf{sk}, x) \mapsto (y, \pi), (\mathsf{pk}, x, y, \pi) \mapsto b$ seen during the execution. Branching then only considers values admitting such globally consistent secure VRF-families as follows after first building a closue of oracle characterization:

- If $(\mathsf{pk}, x, y, \pi) \mapsto 1$ then add $\forall(\mathsf{sk}, \mathsf{pk}), (\mathsf{sk}, x) \mapsto (y, \pi)$
- If $(\mathsf{sk}, x) \mapsto (y, \pi)$ then add $\forall(\mathsf{sk}, \mathsf{pk}), (\mathsf{pk}, x, y, \pi) \mapsto 1$

- For $\mathsf{Gen}_{\mathsf{VRF}}$: we branch
- For $\mathsf{Eval}_{\mathsf{VRF}}$:
- For $\mathsf{Vfy}_{\mathsf{VRF}}$:

The idea. The obstruction to extending the proof to the $(n-1)$-out-of-n setting is that arbitrary combinations of obscure-query answers need not correspond to any globally valid execution of the VRF oracles. In particular, after independently branching over obscure queries to two secure oracles $\mathcal{O}_j, \mathcal{O}_k$, the resulting mixed assignment may violate either perfect correctness or unique provability and therefore fail to correspond to any realizable secure VRF-family.

To avoid this issue, we restrict branching to assignments that are *globally realizable*. Intuitively, every branch considered by the simulator must admit a completion into a family of secure VRF oracles satisfying perfect correctness and unique provability.

Formally, let $\mathcal{T}$ denote the partial transcript generated during the execution. The simulator maintains the following sets:

$$\begin{aligned}\mathcal{K} &:= \{(\mathsf{sk}, \mathsf{pk})\},\\ \mathcal{E} &:= \{((\mathsf{sk}, x), (y, \pi))\},\\ \mathcal{V} &:= \{((\mathsf{pk}, x, y, \pi), b)\}.\end{aligned}$$

Here:

- $(\mathsf{sk}, \mathsf{pk}) \in \mathcal{K}$ records generated key pairs,
- $((\mathsf{sk}, x), (y, \pi)) \in \mathcal{E}$ records evaluation queries,
- $((\mathsf{pk}, x, y, \pi), b) \in \mathcal{V}$ records verification queries.

We say that a partial transcript $\mathcal{T}$ is *globally consistent* if it admits an extension to a family of secure VRF oracles satisfying:

1. perfect correctness,
2. unique provability,
3. consistency of repeated oracle queries.

Equivalently, $\mathcal{T}$ must satisfy the closure conditions induced by VRF correctness:

- If $((\mathsf{pk}, x, y, \pi), 1) \in \mathcal{V}$, then for every $(\mathsf{sk}, \mathsf{pk}) \in \mathcal{K}$ we must have $((\mathsf{sk}, x), (y, \pi)) \in \mathcal{E}$.
- If $((\mathsf{sk}, x), (y, \pi)) \in \mathcal{E}$, then for every $(\mathsf{sk}, \mathsf{pk}) \in \mathcal{K}$ we must have $((\mathsf{pk}, x, y, \pi), 1) \in \mathcal{V}$.

Branching is then restricted to assignments preserving global consistency.

Under this restriction, every obscure-query assignment considered by the proof corresponds to a realizable secure VRF-family. Hence the space of admissible assignments becomes closed under single-coordinate replacement. Consequently, oracle-wise joint irrelevance implies full joint irrelevance by a standard telescoping argument over the coordinates.

Assume there exist two admissible globally consistent assignments $\mathbf{y}, \mathbf{z}$ such that

$$f_t(\mathbf{C}, \mathbf{y}) \neq f_t(\mathbf{C}, \mathbf{z}).$$

Since the admissible assignment space is closed under single-coordinate replacement, one may interpolate between $\mathbf{y}$ and $\mathbf{z}$ by changing one oracle-coordinate at a time. Oracle-wise joint irrelevance implies that the value of f_t remains invariant at every step, yielding a contradiction.